



Exploring Jet Substructure in Public LHC Data

Alexandre Maria Neves de Barbosa Mendonça

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Orientador(es)

Professor Doutor Nome Completo

Júri

Presidente: Professor Doutor Nome Completo

Orientador: Professor Doutor Nome Completo

Vogais: Professor Doutor Nome Completo
Doutor Nome Completo

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AGRADECIMENTOS

RESUMO

Palavras-chave:

ABSTRACT

Keywords:

LIST OF ABBREVIATIONS AND SYMBOLS

1 INTRODUCTION

1.1 Context and motivation

The Large Hadron Collider (LHC) at CERN delivers the highest energy proton-proton collisions ever produced in a laboratory and, with them, an enormous sample of strongly interacting final states. The majority of these collisions are governed by Quantum Chromodynamics (QCD), the gauge theory of the strong interaction. The study and comprehension of these collisions are not only a prerequisite for suppressing backgrounds in searches for new phenomena but also the structure of the hadronic final state can be examined and compared to theory directly, at scales where the coupling is weak enough for perturbation theory to be applied and yet the dynamics still remain rich.

The most prominent objects in these final states are jets. Jets are collimated sprays of hadrons that emerge from the fragmentation of high energy quarks and gluons. A jet is not a fundamental object but instead a definition. A clustering algorithm specifies which detector inputs are grouped together and a recombination scheme specifies how momentum is assigned to a result. This defines a jet, from which an interesting and important question then arises. What does the radiation inside the jets look like? This domain is called jet substructure, which over the past decade has grown from a collection of tagging tools into a precision probe of QCD dynamics in its own right.

In order to organise and look at the internal structure of a jet, one can use the Lund jet plane [1]. The Lund jet plane is a two dimensional map of a jet's branching history on which soft, collinear and non perturbative regions of QCD radiation occupy distinct territories. This map records where the jet's internal emissions occur, but it sets aside an intriguing piece of information, the orientation of each splitting relative to the one before. This orientation is not random. Through the linear polarisation of intermediate gluons, successive splittings are correlated in their relative azimuthal angle, producing a characteristic modulation in the distribution of that angle. This is a genuine quantum interference effect which survives only because the intermediate gluon is described by an amplitude rather than a probability, and it is absent from any picture that treats successive emissions as independent classical branchings.

These collinear spin correlations are well understood theoretically and are implemented in modern parton shower algorithms, yet establishing them cleanly in collision data has proved rather difficult. The reason is that the effect carries opposite signs in different splitting channels. The $g \rightarrow q\bar{q}$ and $g \rightarrow gg$ channels contribute with opposite sign, so a measurement that sums all channels sees the two contributions partially cancel. This interference is not absent from such an inclusive measurement. It is diluted and the apparent size reflects the accidental flavour composition of the sample rather than the strength of the underlying physics.

This is the obstacle the present work addresses. A meaningful measurement of the effect requires knowing which splitting produced each emission. However, the flavour of an individual splitting inside a jet is not directly observable and must be inferred. Working entirely on public CERN Open Data [2], more precisely with CMS Open Data [3], this dissertation builds a reproducible pipeline from the released samples through Lund declustering to the azimuthal spin observable, and introduces a flavour selection that isolates the $g \rightarrow q\bar{q}$ channel. The resulting modulation is reported not as a single number but as a function of channel purity achieved by that selection, a presentation that separates the physics of the

interference from the performance of the selection and allows the magnitude of the effect to be read off in the limit of a pure sample.

The Lund declustering angle $\Delta\psi_{12}$ is not the only way to reach this effect. The same $\cos 2\psi$ modulation also imprints itself on the flow of energy within a jet, and can be accessed through a three-point energy correlator, which probes the relative azimuthal orientation of energy deposited by triplets of particles without reference to any clustering history. The two observables are complementary: one reads the effect from the reconstructed branching sequence, the other from the energy flow directly, and each carries a different exposure to the choices and imperfections of the measurement. This work develops both, treating the Lund observable as the primary measurement and the energy correlator as an independent cross-check on the same underlying physics.

The fact that this analysis rests on public data is not incidental. The CERN Open Data Portal [2] has made it possible for work outside the experimental collaborations to produce meaningful results and to carry out genuinely new measurements. The present work is intended as a contribution in that spirit. It builds an end to end pipeline from a single CMS dataset to the final measurements, establishing along the way a workflow for accessing, handling and measuring with CMS Open Data.

1.2 Objectives

The main aim of this dissertation is to measure collinear spin correlations in QCD jets using public LHC data and to establish the flavour selection that such a measurement requires in order to be meaningful. For the purpose of achieving that goal, the work is organised around a sequence of objectives. The first is to build a reproducible pipeline over CERN Open Data samples capable of extracting the necessary variables to begin a jet substructure analysis. This means that the data samples must contain jet constituent information, thus the data format chosen must retain that level of detail, ruling out the more compact samples in favour of one that preserves per-constituent quantities.

With that pipeline in place, the second objective is to construct the primary and secondary Lund jet planes and to verify that they reproduce the known features of QCD radiation, since this agreement is what establishes that the declustering is sound before any finer observable is built on top of it. Following that, the third objective is to implement the azimuthal observable $\Delta\psi_{12}$ between the hardest primary and secondary branchings, and to study its distribution both with and without grooming, recovering the inclusive result before attempting to sharpen it.

The fourth objective is the central one, and the point at which this work departs from a purely inclusive analysis. Because the spin-interference signal carries opposite signs in the $g \rightarrow q\bar{q}$ and $g \rightarrow gg$ channels, an inclusive measurement dilutes it. Isolating a single channel is therefore not a refinement but a precondition for measuring the effect at its true strength. This objective is thus to develop a flavour selection capable of enriching the $g \rightarrow q\bar{q}$ channel, and to quantify both how well it performs and how pure a sample it can reach in principle. The fifth objective builds directly on it. It is to measure $\langle \cos 2\psi_{12} \rangle$ as a function of the channel purity delivered by that selection and to establish the value characteristic of a pure $g \rightarrow q\bar{q}$ sample.

Finally, the work aims to assess the systematic uncertainties affecting this measurement and to identify the dominant limitations of the approach, so that the extensions it opens up can be set out on solid ground.

1.3 Thesis outline

Chapter 2 develops the theoretical material on which the measurement rests, such as the elements of QCD relevant to jet formation, the definition and reconstruction of jets and the algorithms used to build and decluster them, the Lund jet plane, and the origin of collinear spin correlations together with the two observables through which this work accesses them. Chapter 3 turns to the data and the analysis, and covers the CERN Open Data resources and the CMS detector, the data format and dataset used, and the pipeline that runs from the released samples through Lund declustering to the azimuthal observable, the flavour selection, and the treatment of systematic uncertainties. Chapter 4 presents the validation of the pipeline and the measurement itself, and discusses the results in the light of the expectations set out in Chapter 2. Finally, Chapter 5 returns to the objectives of Section 1.2, states what has been established and with what confidence, and outlines the extensions that the present framework makes possible.

2 THEORETICAL BACKGROUND

2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the quantum field theory of the strong interaction, a non-Abelian gauge theory built on the colour symmetry group $SU(3)$ [4]. Its fundamental degrees of freedom are quarks, which carry colour charge, and gluons, the gauge bosons that mediate the interaction. The non-Abelian character of the theory has a decisive consequence absent from Quantum Electrodynamics. The gluons themselves carry colour, and therefore couple directly to one another. To a large extent, the dynamics of the strong interaction are the dynamics of this gluon self-coupling, and the two properties of QCD that govern everything in this chapter both follow from it.

2.1.1 Asymptotic freedom and confinement

The first property is asymptotic freedom. The strong coupling α_s decreases as the momentum transfer of an interaction increases, so that at the high scales probed in LHC collisions the quarks and gluons interact weakly and perturbation theory applies [5, 6]. The running of the coupling is captured at leading order by

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2) \frac{33 - 2n_f}{12\pi} \ln \frac{Q^2}{\mu^2}}, \quad (2.1)$$

where Q is the scale of the process, μ a reference scale, and n_f the number of active quark flavours. The positive sign of the coefficient $33 - 2n_f$, which holds for any physical number of flavours, is what makes the coupling fall at large Q . It is a direct imprint of the gluon self-interaction, and its discovery established that a perturbative treatment of hard processes is justified. It is convenient to trade the reference scale for a single dimensionful constant Λ_{QCD} , the scale at which the perturbative coupling formally diverges, of an order of a few hundred MeV. In terms of it the coupling is small for $Q \gg \Lambda_{\text{QCD}}$ and grows strong as Q approaches it.

This growth is the second property called confinement. At low momentum transfer α_s becomes large and colour-charged particles cannot be isolated. A quark or gluon produced in a hard scattering does not propagate freely to a detector, because as it separates from other colour charges, the energy stored in the colour field between them grows with distance until it becomes energetically favourable to create a new quark-antiquark pairs from the vacuum. The original parton is thereby dressed into a spray of colour-neutral hadrons. This is why the strongly interacting final state reaches the detector not as partons but as the collimated sprays introduced in Chapter 1, and why any comparison between a perturbative calculation and a measurement must bridge the divide between the partonic and hadronic descriptions.

2.1.2 Soft and collinear radiation

The bridge between the hard parton and the observed hadrons is built from successive emissions, and the pattern of those emissions is far from uniform. When a parton radiates a gluon, the emission probability

at first order in the coupling takes the schematic form

$$d\mathcal{P} \propto \frac{2\alpha_s C}{\pi} \frac{d\theta}{\theta} \frac{dE}{E}, \quad (2.2)$$

where E and θ are the energy and emission angle of the radiated gluon and C is a colour factor. The two factors carry the characteristic structure of QCD radiation. The factor $d\theta/\theta$ diverges as $\theta \rightarrow 0$ meaning that emissions are enhanced when the gluon is radiated close to the direction of its emitter. This is known as the collinear divergence. The factor dE/E diverges as $E \rightarrow 0$ meaning that emissions are enhanced when the gluon is soft. This is known as the soft divergence. Together they mean that a hard parton radiates predominantly softly and at small angles, and that number of such emissions is, in a precise logarithmic sense, large.

The collinear structure is described more completely by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) splitting functions [4], which give the probability that a parton of one type splits into two, as a function of the momentum fraction z carried by one of the products. For the three splittings relevant to this work they read, at leading order,

$$P_{q \rightarrow qg}(z) = C_F \frac{1+z^2}{1-z}, \quad (2.3)$$

$$P_{g \rightarrow gg}(z) = C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z) \right], \quad (2.4)$$

$$P_{g \rightarrow q\bar{q}}(z) = T_R [z^2 + (1-z)^2], \quad (2.5)$$

with colour factors $C_F = 4/3$, $C_A = 3$ and $T_R = 1/2$. The distinction between the gluon splitting channels of Eqs. (2.4) and (2.5), one producing a pair of gluons and the other a quark-antiquark pair, will prove central. The two channels differ not only in these rates but in the spin structure of their emissions, and it is that difference which the spin observables of Section 2.4 are designed to probe.

These divergences are not a pathology. They are integrable, and in any physical, infrared-safe observable the divergent real emissions cancel against corresponding virtual corrections, leaving a finite result. This cancellation is the requirement that shapes the very definition of a jet in Section 2.2. Their logarithmic structure, however, is physical and measurable, and it is exactly the structure that the Lund jet plane of Section 2.3 is built to expose. By mapping each emission onto logarithmic axes of angle and transverse momentum, it turns the double logarithmic enhancement of Eq. (2.2) into an approximately uniform density, against which finer correlations stand out.

2.1.3 Parton showers and hadronization

Because a single hard parton typically radiates many times before hadronizing, the emissions can be organised into a sequence ordered from the hardest, widest-angle splittings down to the soft and collinear ones. This sequence is what is known as a parton shower. Modern Monte Carlo event generators implement this sequence probabilistically through the splitting functions of Eqs. (2.3) and (2.5), evolving each parton down from the hard scale of the collision to a low scale of order 1 GeV, at which the perturbative description breaks down and confinement takes over. Below that scale the partons are converted into hadrons by phenomenological hadronization models, tuned to data rather than derived from first principles.

This picture, a hard, perturbatively calculable core, dressed by a shower of increasingly soft and collinear emissions and finally hadronized, underlies the whole of the analysis that follows. The observables studied in this work are built to read the perturbative shower through the hadrons that reach the detector, and for this reason the measurement is deliberately confined to a hard kinematic regime, where the perturbative component dominates and non-perturbative corrections are comparatively suppressed.

2.2 Jets

As established in Section 2.1, a coloured parton produced in a hard scattering does not reach the detector directly. It radiates and hadronizes, arriving as a collimated spray of hadrons. A jet is the operational device that inverts this process approximately, grouping those hadrons back together so that their combined momentum estimates that of the parton that produced them. A jet is therefore not a physical object with a sharp definition but a construction, and its properties depend on the prescription used to build it. This section sets out that prescription: the requirement any jet definition must satisfy to be usable in perturbative QCD, and the three algorithms, built on that requirement, on which this work relies.

2.2.1 Jet algorithms and infrared–collinear safety

Two ingredients specify how a jet is built. A jet algorithm decides which inputs, from detector-level energy deposits, reconstructed particles, or, in a calculation, partons, are grouped together into a jet. A recombination scheme specifies how the momenta of those grouped inputs are combined into the momentum of the jet. Throughout this work the inputs are the particle-flow candidates described in Chapter 3, and momenta are combined in the E -scheme, in which four-momenta are simply added. This scheme is Lorentz invariant and yields jets with non-zero mass.

Unfortunately, not every conceivable grouping prescription is admissible. The requirement that disqualifies most of them is infrared and collinear (IRC) safety. An observable is infrared safe if its value is unchanged by the addition of an arbitrarily soft emission, and collinear safe if it is unchanged when one of its inputs is replaced by two collinear inputs sharing its momentum. A jet algorithm is IRC safe when the set of jets it returns is stable under both these operations. A soft particle must not create or destroy a jet and a collinear splitting must not change how particles are assigned to jets.

The reason why this requirement is not optional follows directly from Section 2.1. The emission probability of Eq. (2.2) diverges in both the soft and the collinear limits. In any IRC-safe observable these divergent real emissions cancel against equally divergent virtual corrections, leaving a finite result. In an IRC-unsafe observable, that cancellation is spoiled and a fixed-order perturbative prediction returns infinity rather than a number. An IRC-unsafe jet definition is therefore not merely inconvenient but it also cannot be compared with perturbative QCD at all. Since the purpose of measuring jets is to confront the theory with data, IRC safety is a precondition and not a refinement.

This criterion has been decisive in the historical development of jet algorithms. The earliest definitions were cone algorithms, which grouped particles falling within a fixed angular radius of a candidate axis. In their simplest, seed-based forms these algorithms were not IRC safe because a soft particle could seed a new candidate cone, or a collinear splitting could change which particles acted as seeds, and either could alter the resulting jets. Much effort was devoted to repairing them, and eventually producing seedless variants that restored collinear safety. The alternative family, the sequential recombination algorithms, are

IRC safe by construction, and it is to this family that algorithm used in this work belongs.

It is worth making concrete how such a failure arises, since the same mechanism will reappear when flavour is assigned to jets in Section ???. Consider a seed-based cone algorithm applied to an event with two hard particles whose separation is slightly greater than the cone radius, so that each seeds its own jet. The addition of a single soft particle between them may seed a third cone that overlaps both, merging the two hard particles into one jet where before there were two which, again, it is an infrared-unsafe change, since an arbitrarily soft emission has altered the hard jets. A collinear splitting produces the collinear-unsafe failure. Replacing one hard seed by two collinear particles, each carrying half its transverse momentum, may leave neither above the seed threshold, so that a jet which existed before the splitting is absent after it. In both cases, the divergent emissions of Eq. (2.2) feed directly into the jets themselves, and the perturbative prediction diverges with them.

The sequential recombination algorithms introduced below evade both failures by never relying on seeds or on fixed geometric cones. They build jets by repeatedly merging inputs according to a distance measure, so that soft or collinear addition changes only the order in which already nearby inputs are combined, never the final assignment of hard momentum to jets. This structural difference is the reason the entire family is IRC safe, and it is what recommends it over the cone algorithms for any measurement to be compared with theory.

A sequential recombination algorithm is defined by a distance measure between each pair of inputs and between each input and the beam. The generalised- k_T family [7] writes these as

$$d_{ij} = \min\left(p_{T,i}^{2p}, p_{T,j}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad (2.6)$$

$$d_{iB} = p_{T,i}^{2p}, \quad (2.7)$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared angular separation of the inputs in rapidity y and azimuth ϕ , R is the jet radius parameter, and the integer p selects the algorithm. At each step the algorithm finds the smallest of all the distances. If it is a d_{ij} , the inputs i and j are recombined into a single input. If it is a d_{iB} , the input i is declared a final jet and removed. The procedure repeats until no inputs remain. The set of jets returned is IRC safe for any integer value of p , but the value of p changes profoundly how the clustering proceeds. The three choices $p = -1$, $p = 0$ and $p = +1$ give, respectively, the anti- k_T , Cambridge/Aachen and k_T algorithms, of which the first two are the tools this work relies on and are described in the following subsections.

Common to all three is the radius parameter R , which sets the angular reach of the clustering through the ratio $\Delta R_{ij}^2/R^2$ in Eq. (2.6). A smaller R yields narrower jets that resolve nearby structure but capture less of a single parton's radiation. A larger R collects more of that radiation at the price of admitting more contamination from unrelated soft activity. The appropriate choice depends on the use to which the jets are put, and the two algorithms this work relies on are used with different values of R for correspondingly different purposes. This point is, again, taken up in Sections 2.2.2 and 2.2.3.

2.2.2 The anti- k_T algorithm

The choice $p = -1$ in Eqs. (2.6) and (2.7) defines the anti- k_T algorithm [8]. With this value the inter-particle distance becomes

$$d_{ij} = \min\left(\frac{1}{p_{T,i}^2}, \frac{1}{p_{T,j}^2}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad (2.8)$$

so that the distance scales with the inverse squared transverse momenta of the inputs. The smallest distances, and therefore the first recombinations, are those involving the hardest particles. The clustering is consequently driven by hard inputs. A hard particle accretes the softer inputs around it before soft inputs cluster among themselves.

The name of the algorithm reflects this inversion. In the k_T algorithm, the choice $p = 1$, the distance scales with the positive powers of transverse momentum, so that soft inputs are clustered first and the jet is assembled from the bottom up, following the inverse of the physical showering order. The anti- k_T algorithm reverses this. Taking $p = -1$ turns the soft-first ordering of k_T into a hard-first one. It is, in effect, the k_T algorithm run with the role of soft and hard interchanged, and the prefix “anti” records precisely that inversion. The two sit at opposite ends of the generalised family, with Cambridge/Aachen, insensitive to transverse momentum altogether, between them.

This ordering has a striking geometric consequence. A hard particle with no other hard particle within an angular distance $2R$ collects all softer inputs within a distance R of it, forming a jet that is exactly circular in the (y, ϕ) plane. Where two hard particles lie closer than $2R$, the boundary between their jets is set by their relative transverse momenta, the harder one retaining the larger share, but the jets remain regular in outline. Soft radiation, precisely because it enters the clustering last, cannot deform these boundaries. It is distributed among the hard jets without reshaping them.

A useful way to characterise this stability is through the notion of jet area, the region of the (y, ϕ) plane from which a jet gathers its inputs. This area may be probed operationally by adding a large number of infinitesimally soft “ghost” particles to an event and recording which jet each is assigned to. For most algorithms the resulting areas are irregular and fluctuate from jet to jet, because soft particles influence the clustering. However, for the anti- k_T algorithm, where soft particles are clustered last and cannot reshape the hard jets, the areas are close to the ideal value πR^2 and vary little. A stable, predictable area is what makes the subtraction of pile-up and underlying-event contamination tractable, since that contamination scales with the area over which each jet collects it.

It is this regularity that has made the anti- k_T algorithm the standard jet definition at the LHC. Because the jet boundary is stable against soft radiation, the area over which a jet collects contamination from pile-up and the underlying event is well defined and nearly constant, and the corresponding energy corrections are straightforward to calibrate. The algorithm thus combines the IRC safety guaranteed for the whole generalised- k_T family with a jet shape that is simple to treat experimentally, a combination that its two siblings, whose boundaries are irregular, do not offer.

The prominence of the algorithm is also a matter of timing. When the LHC began taking data, a jet definition was needed that was both IRC safe, so that measurements could be compared with the increasingly precise perturbative calculations then becoming available, and computationally fast enough to run on the very large event samples the machine would produce. The anti- k_T algorithm, together with its efficient implementation in the FastJet package [9], met both needs, and it was adopted as the baseline jet

definition by the LHC experiments, a role it has held since.

For these reasons the anti- k_T algorithm was used in this work to define the jets, with radius parameter $R = 0.8$. Jets constructed in this way are conventionally denoted as AK8. What the algorithm does not provide, however, is a physically meaningful account of how the jet was assembled. Because its sequence is ordered by hardness rather than by angle, the intermediate recombinations bear no correspondence to the sequence of physical splittings in a parton shower, which is ordered in angle. The clustering history of an anti- k_T jet is, in this sense, an artefact of the algorithm rather than a record of the underlying radiation. Recovering that history, which is what the substructure observables of this work require, calls for a different member of the family, applied to the constituents of the jet once it has been defined. That algorithm is the Cambridge/Aachen algorithm and it is the subject of the next subsection.

2.2.3 The Cambridge/Aachen algorithm

2.3 The Lund jet plane

2.4 Collinear spin correlations

3 METHODS

4 RESULTS AND DISCUSSION

5 CONCLUSIONS

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